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**BIT 2324 GEOGRAPHICAL INFORMATION
SYSTEMS**

LAST REVISION ON March 1, 2016





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke



LESSON 8

Basics of Remote Sensing

Learning outcomes

By the end of this topic you should be able to;

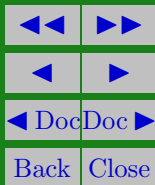
- *Explain the functions of a firewall*
- *Explain the limitations of firewalls*
- *Describe various types of firewalls*



8.1. Introduction

No article on GIS would be complete without mentioning some of the applications of spatial data. One important application is wildlife habitat assessment. Spatial data can be used to calculate the home range or territory of a particular species. They can also be used to eliminate areas where a certain animal would rarely or never be found (e.g., slope data could be used to eliminate extremely steep slopes, or elevation data could be used to eliminate elevations above a certain level). In addition, by simply moving a 3 x 3 pixel window over the vegetation map, calculations

of juxtaposition and interspersions can be made. Juxtaposition can measure the number of different vegetation types in a certain area, while interspersions measures the value of the edges.





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Both are very important to wildlife, since food and cover must be in close proximity in order for the habitat to be useful. Obviously, other proximity analyses are possible, such as distances to water and roads. A wildlife habitat assessment model, typically based on knowledge compiled by one or more wildlife specialists, combines the spatial data, its derivatives, and other information necessary to predict habitat quality. The more important variables are frequently given higher weights. This analysis can determine areas of good, marginal, and poor habitat, providing the first step toward intelligently improving the area.

Another application, predicting snowmelt runoff from a watershed, involves a statistical model using regression analysis. As in the wildlife example, the same spatial data are useful. However, in this case the regression analysis chooses the important



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variables in stead of the wildlife biologist. In predictions now melt runoff the important parameters are the snow covered area and the snow-water equivalent. Other spatial data could include elevation, slope, aspect, vegetation, thermal emittance, near-infrared reflectance, and soils. In addition, data must be collected at specific sample locations on the ground. All these data are tested by the regression analysis to see which are significant in predicting runoff; the best model has the highest correlation coefficient.

This model is then used to predict snowmelt runoff over the entire watershed. These two examples demonstrated different modeling approaches using spatial data. It should not be inferred from these examples that GIS only applies to natural resources. GIS has many applications, from aiding decisions to evaluating



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potted owl habitat. T hoses tartingto uses patial data as a tool for decisionmaking should review the literature in their particular field and learn as much as possible from pasts uccesseasn df ailures. Finally, a warning. A GIS is a very powerfulecision-makintgo ol. However, it is just a tool. There is a great temptation to leant oo heavilyo n the computer and to stop thinking. There is also a tendency to collect too much spatial data. One more layer of data is not always the answer. Be thoughtful and wary, and GIS will serve you well.

8.2. GIS Applications

- Site selection
 - Helicopter Landing Zones
 - Amphibious Assault (Water Depth)



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- Buffer Zones
- Flight Planning
- Battlefield Visualisation
- Cross country movement
 - Route planning
 - Intervisibility study
- Facilities management
- Airfield assessment
- Road network analysis (convoys)
- Propagation coverages
- Observation post siting analysis
- Perspective views





Revision Questions

EXERCISE 1.  Explain security attacks that can be implemented on packet-filtering routers & countermeasures.

Example . List the characteristics of bastion host firewall.

Solution: a) execution file i) have each proxy run as a non-privileged user in a private and secured directory.

□

EXERCISE 2.  Explain the feature of a distributed firewall that distinguishes it from other types of firewalls.



Solutions to Exercises

Exercise 1. IP address spoofing: where intruder transmits packets from the outside with interimum fragment size to include full header.

Exercise 1



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Exercise 2. An important aspect of a distributed firewall configuration is security monitoring. Such monitoring typically includes log aggregation and analysis, firewall statistics, and fine-grained remote monitoring of individual hosts if needed.

Exercise 2